Project-1

First, we have to import this CSV file into Pandas library.Pandas is a powerful data analysis library providing flexible way to do data manipulation.It also provides different features for doing mathematical and scientific calculations.

In this exercise, we will use some of these features. To load the CSV file using Pandas, we need to follow the steps given below.

```
#load CSV file
df=pd.read_csv('railway_gauges 1.csv')
df.head()
```

When we load data using Pandas, it is returned as a DataFrame object. Using this DataFrame, we can view the data easily. The head() function is used to display the first 5 rows, and the tail() function is used to display the last 5 rows of the CSV file.

In the above example, we used df.head(), which shows the first 5 rows of the loaded CSV file.

Year	Broad Gauge	Metre Gauge	Narrow Gauge	Total
1964-65	3521	2891	464	6876
1965-66	3603	2913	470	6986
1966-67	3615	2911	481	7007
1967-68	3601	2920	506	7027
1968-69	3603	2921	508	7032

Below are some of the essential Pandas functions and attributes.

- **iloc [ ]** — is primarily an integer based location search — iloc[3]
- **loc[]** — is primarily a label based location search — loc['Broad Gauge']
- **idxmax()-** is a function to identify the index of first occurrence of maximum over request axis.

Using idxmax() function:

```
#Find which year has the maximum installation
df.iloc[[df['Total'].idxmax()]]
```

	Year	Broad Gauge	Metre Gauge	Narrow Gauge	Total
48	2012-13	6577	379	216	7172

Data Visualisation:

We can use Matplotlib libraries to represent the data in a graphical way. This helps us understand the data more clearly. In our example, we use the following steps to display the data using a bar chart.

```
#plot Data using bar chart
df=df.drop('Total',axis=1)
ax=df.plot(x="Year",kind="bar")
plt.xticks(rotation=70)
plt.xlabel("Year")
plt.ylabel("Total")
plt.title("Gauges: Number of railway tracks installed per year")
plt.savefig("rail_gauges.png")
plt.show()
```

And the output graph look likes in below.